

Polisetty Naga Chaitanya

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Professional Summary

Electronics and Communication Engineering graduate with a strong foundation in Python programming, full-stack web development, and machine learning. Applied Python, NumPy, Pandas, Scikit-learn, TensorFlow, and OpenCV across academic projects involving hyperspectral image classification, data preprocessing, feature extraction, dimensionality reduction using PCA, and model evaluation. Technical foundation includes Django, SQL, MySQL, HTML, CSS, JavaScript, REST APIs, Git, debugging, testing, object-oriented programming, and problem-solving.

Education

B.Tech – Electronics and Communication Engineering Vel Tech Rangarajan Dr. Sagunthala R & D Institute of Science and Technology, Chennai CGPA: 7.5/10	2022 – 2026
Class XII – MPC Sri Chaitanya Junior College Percentage: 79.2%	2020 – 2022
Class X Balavikas English Medium High School Percentage: 94%	2020

Technical Skills

Programming Languages: Python, SQL, JavaScript, HTML, CSS
Ece core skills: VLSI Design, Internet of Things, Embedded Systems, , Computer Networks
Frameworks and Libraries: Django, Pandas, NumPy, Scikit-learn
Databases: MySQL, Relational Databases, SQL Queries
Data Analysis: Microsoft Excel, Power BI, Data Processing, Data Visualization
Software Development: Web Application Development, Debugging, Testing
Tools: Git, GitHub, Visual Studio Code

Certifications

Python Using AI Workshop - Cognitve Class.ai .	Sep 2025
Operating systems - Infosys Spring Board.	Mar 2025
Sustainable Development and Energy – NPTEL.	Feb 2025
Intelligent Embedded Systems: AI for real-world applications..	Jan 2026
Responsible and safe AI systems – NPTEL	Oct 2025

Project Experience

A Spectral-Spatial Attention Transformer Framework for Robust Hyperspectral Image Classification 2026	Jan 2026 – May
Technologies: Deep learning , computer vision , Python, hyperspectral imaging	
- Developed a spectral-spatial attention transformer model for hyperspectral image classification, leveraging deep learning techniques to extract and analyze spectral-spatial features. - Improved classification accuracy and robustness through attention-based feature learning, enabling effective remote sensing and environmental monitoring applications.	
Bioluminescent Organism Detection in Underwater Images	Jul 2025 – Oct 2025
Technologies: Python, YOLOv5, Computer Vision, Deep Learning, Image Processing	
- Developed an AI-based deep learning system to detect bioluminescent organisms in low-light underwater images using the YOLOv5 object detection algorithm. - Implemented image enhancement, data augmentation, and two-stage training to improve detection accuracy under poor visibility and high-noise conditions.	
Path Optimization and Obstacle Avoidance for an Autonomous Surveillance Robot	Jan 2025 – Apr 2025
Technologies: Python, MATLAB(Simulation), A* Algorithm, Dijkstra's Algorithm, Robotics, Path Planning	
- Developed a path-planning and obstacle-avoidance system for an autonomous surveillance robot using A* and Dijkstra's algorithms. - Simulated multi-Obstacle environments to compute optimal paths, enabling efficient navigation with minimized distance, energy usage and real-time adaptability.	

Personal Attributes

- Quick to adapt to new environments, technologies, and challenges, ensuring efficient problem-solving
- Strong communicator with leadership and organizing abilities. Confidently engages audiences with impactful presentations